

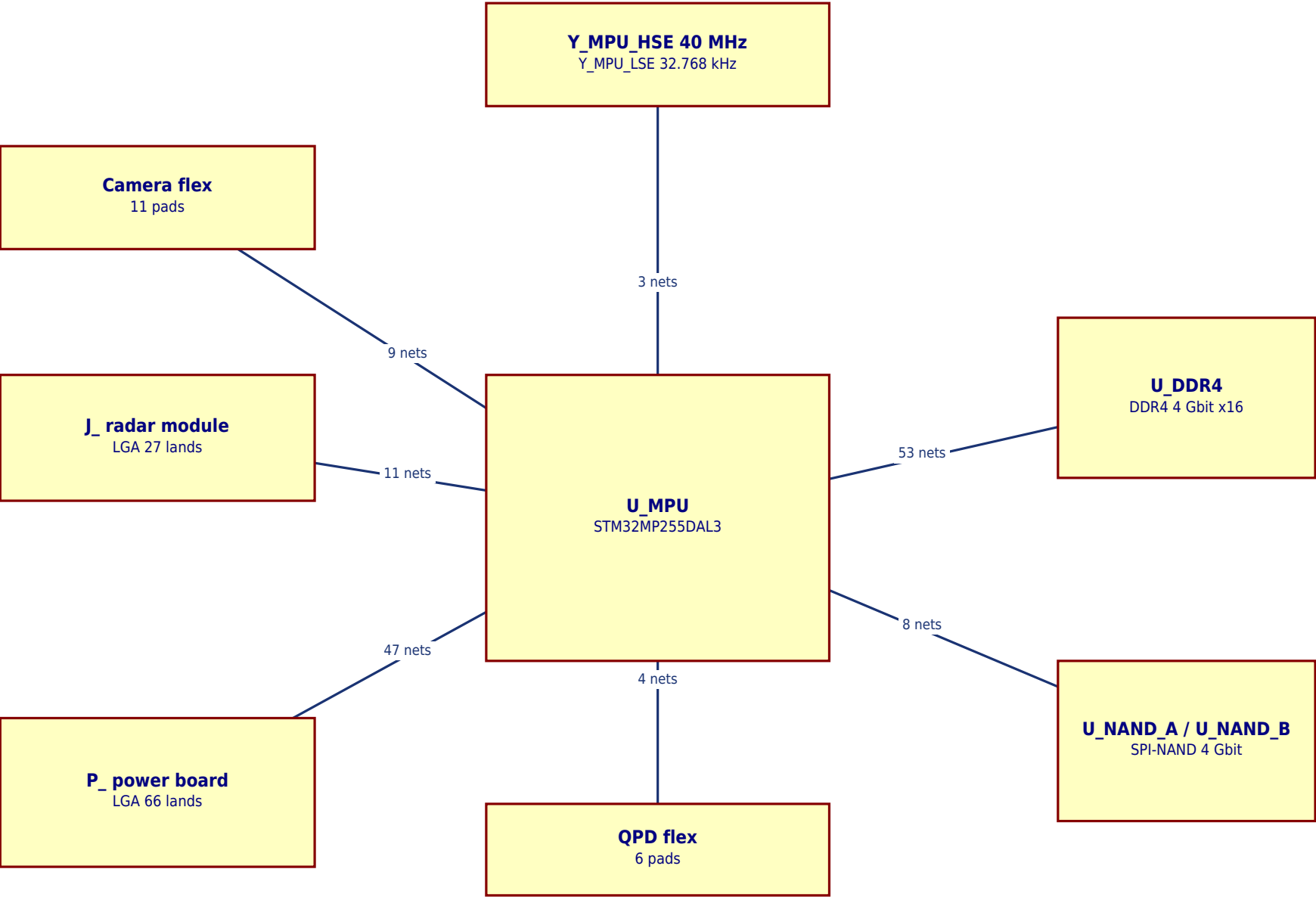
Companion Compute - schematic

STM32MP255 + 4 Gbit DDR4 (x16) + 2 × 4 Gbit SPI-NAND · Ø34 mm, 20 copper layers

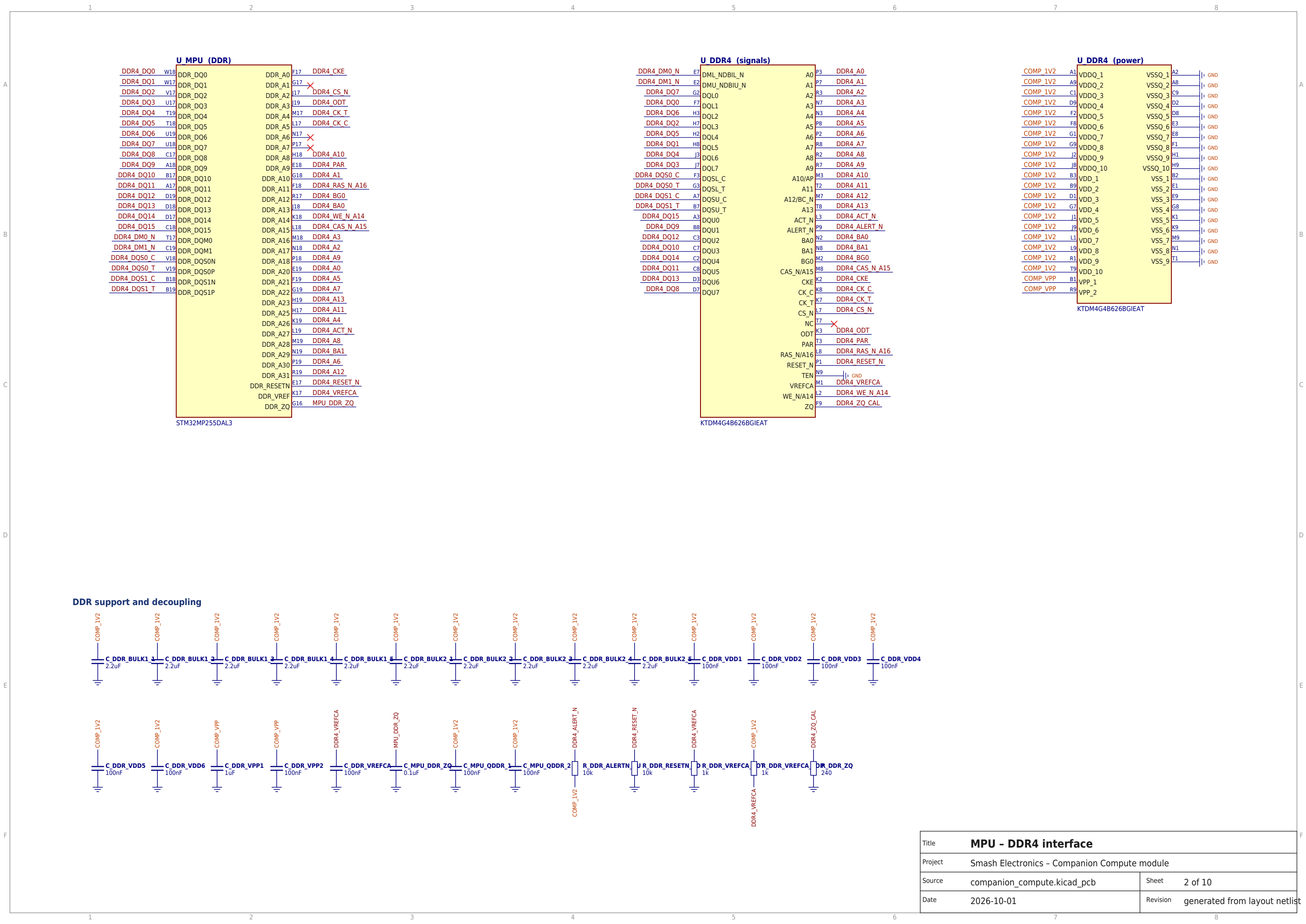
Generated from the layout netlist (companion_compute.kicad_pcb); the smash flow has no schematic capture. Same net name = connected.

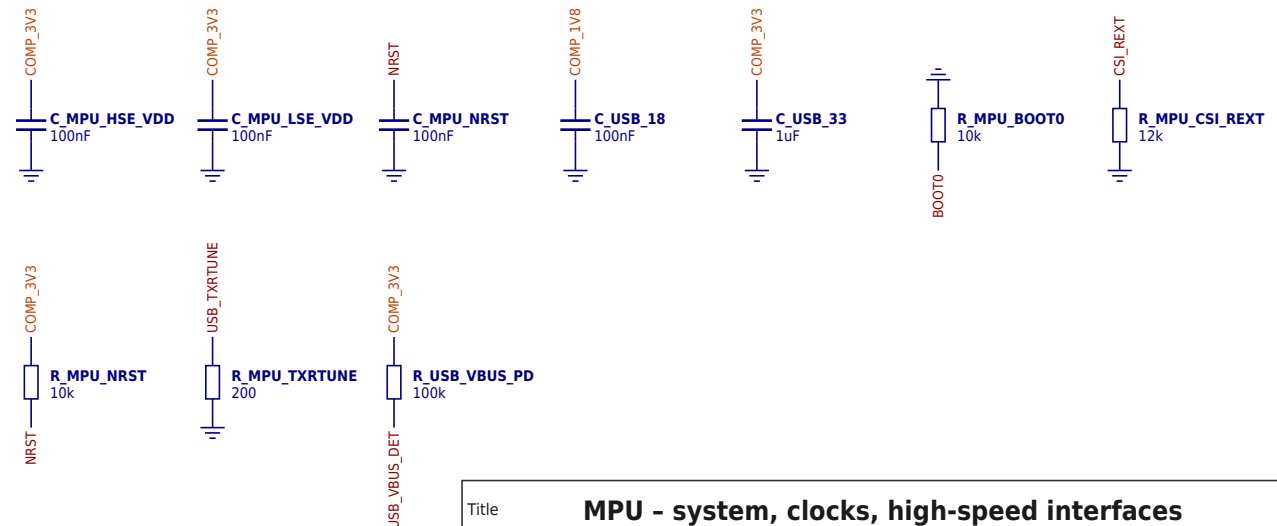
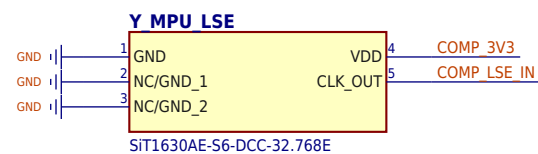
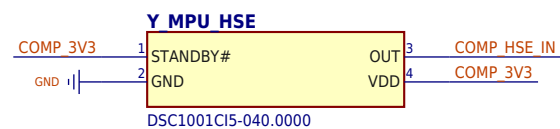
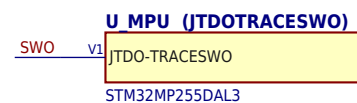
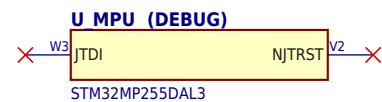
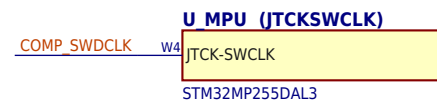
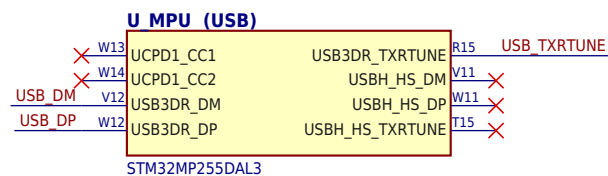
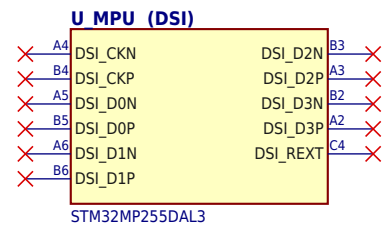
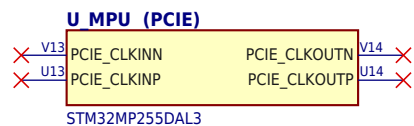
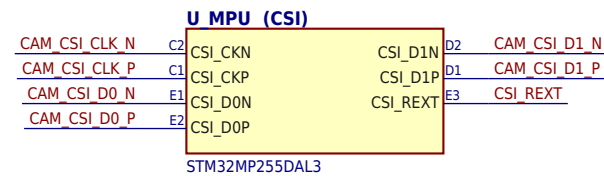
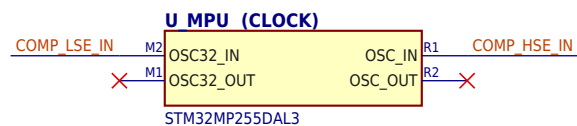
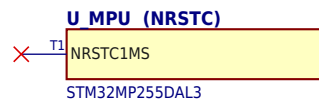
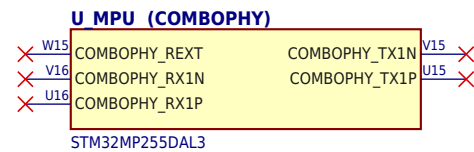
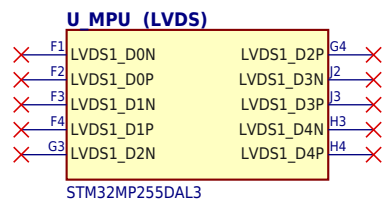
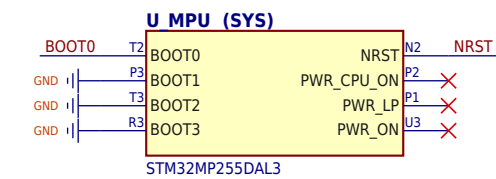
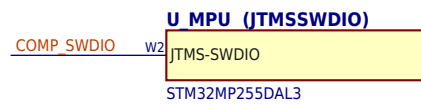
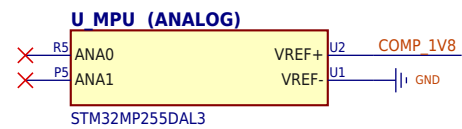
Sheets

- 1 Cover and block diagram
- 2 MPU - DDR4 interface
- 3 MPU - GPIO ports A-E
- 4 MPU - GPIO ports F-Z
- 5 MPU - system, clocks, high-speed interfaces
- 6 MPU - power
- 7 Boot storage - SPI-NAND
- 8 Board interfaces - LGA and flex
- 9 Net cross-reference (1/2)
- 10 Net cross-reference (2/2)

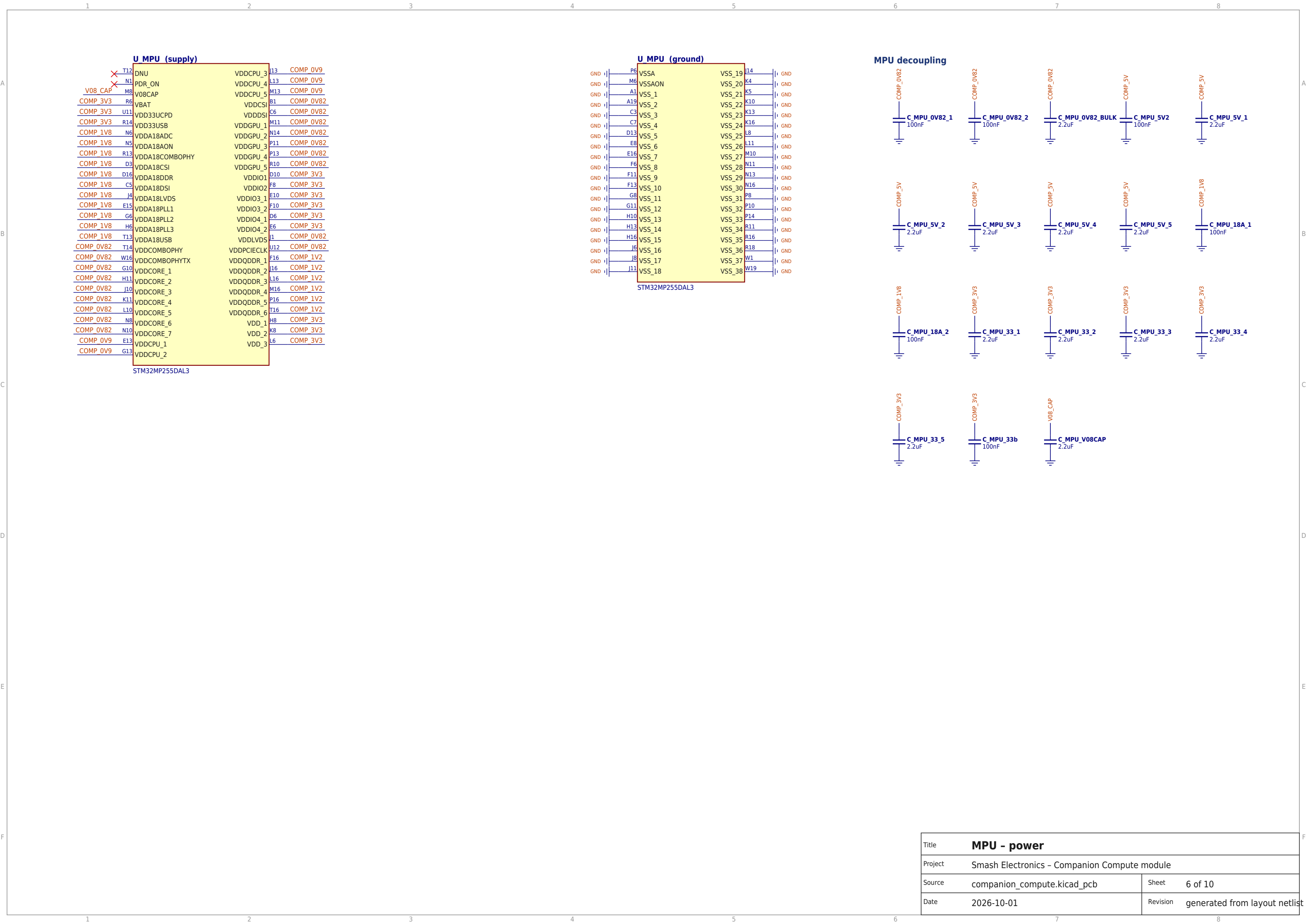


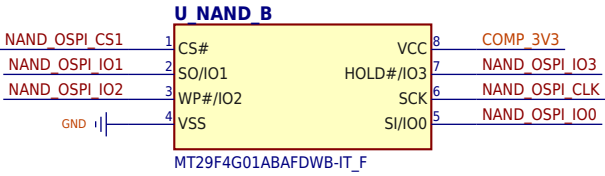
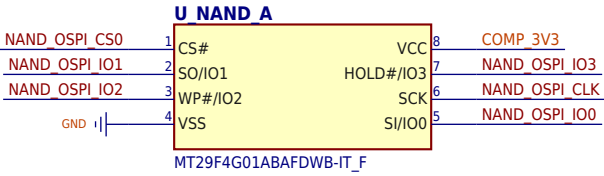
Title	Cover and block diagram		
Project	Smash Electronics - Companion Compute module		
Source	companion_compute.kicad_pcb	Sheet	1 of 10
Date	2026-10-01	Revision	generated from layout netlist



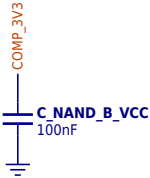
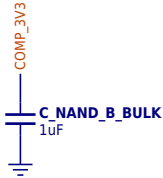
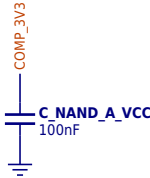
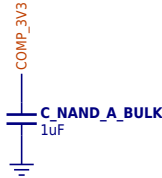


Title	MPU - system, clocks, high-speed interfaces		
Project	Smash Electronics - Companion Compute module		
Source	companion_compute.kicad_pcb	Sheet	5 of 10
Date	2026-10-01	Revision	generated from layout netlist





Decoupling



Both devices share the MPU OCTOSPI/SPI bus; chip selects are separate nets.

Title	Boot storage - SPI-NAND		
Project	Smash Electronics - Companion Compute module		
Source	companion_compute.kicad_pcb	Sheet	7 of 10
Date	2026-10-01	Revision	generated from layout netlist

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		1	2	3	4	5	6	7	8
A	Net		#	Connections					
	WBA_BOOT0		2	P_companion_compute_spacer_power_board_companion_compute.48, U_MPU.B11					
	WBA_HCI_IRQ		2	P_companion_compute_spacer_power_board_companion_compute.49, U_MPU.A8					
	WBA_HCI_MISO		2	P_companion_compute_spacer_power_board_companion_compute.50, U_MPU.E12					
	WBA_HCI_MOSI		2	P_companion_compute_spacer_power_board_companion_compute.51, U_MPU.A7					
	WBA_HCI_NSS		2	P_companion_compute_spacer_power_board_companion_compute.52, U_MPU.A16					
	WBA_HCI_SCK		2	P_companion_compute_spacer_power_board_companion_compute.53, U_MPU.A15					
	WBA_RST_CMD		2	P_companion_compute_spacer_power_board_companion_compute.54, U_MPU.A9					
B	WBA_UART_RX		2	P_companion_compute_spacer_power_board_companion_compute.55, U_MPU.P12					
	WBA_UART_TX		2	P_companion_compute_spacer_power_board_companion_compute.56, U_MPU.M15					
C									
D									
E									
F									

Title		Net cross-reference (2/2)	
Project		Smash Electronics - Companion Compute module	
Source	companion_compute.kicad_pcb	Sheet	10 of 10
Date	2026-10-01	Revision	generated from layout netlist